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```
from abc import ABC, abstractmethod

# -----
# 1. ORDER CLASSES
# -----

class Order(ABC):

    def __init__(self, order_id, amount):
        self.order_id = order_id
        self.amount = amount

    @abstractmethod
    def get_total(self):
        pass

    @abstractmethod
    def get_order_type(self):
        pass

class RegularOrder(Order):

    def get_total(self):
        return self.amount

    def get_order_type(self):
        return "Regular Order"

class DiscountedOrder(Order):

    def __init__(self, order_id, amount, discount):
        super().__init__(order_id, amount)
        self.discount = discount

    def get_total(self):
        return self.amount - self.discount

    def get_order_type(self):
        return "Discounted Order"
```

```
class PriorityOrder(Order):

    def __init__(self, order_id, amount, priority_fee):
        super().__init__(order_id, amount)
        self.priority_fee = priority_fee

    def get_total(self):
        return self.amount + self.priority_fee

    def get_order_type(self):
        return "Priority Order"
```

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# -----
# 2. PAYMENT SYSTEM
# -----
```

```
class PaymentMethod(ABC):
```

```
    @abstractmethod
    def pay(self, amount):
        pass
```

```
class CreditCardPayment(PaymentMethod):
```

```
    def pay(self, amount):
        print(f"Paid {amount} using Credit Card")
```

```
class UPIPayment(PaymentMethod):
```

```
    def pay(self, amount):
        print(f"Paid {amount} using UPI")
```

```
class WalletPayment(PaymentMethod):
```

```
    def pay(self, amount):
        print(f"Paid {amount} using Wallet")
```

```
# -----
# 3. NOTIFICATION SYSTEM
# -----

class Notification(ABC):

    @abstractmethod
    def send(self, message):
        pass

class EmailNotification(Notification):

    def send(self, message):
        print(f"Email sent: {message}")

class SMSNotification(Notification):

    def send(self, message):
        print(f"SMS sent: {message}")

class PushNotification(Notification):

    def send(self, message):
        print(f"Push Notification: {message}")

# -----
# 4. STORAGE SYSTEM
# -----

class Storage(ABC):

    @abstractmethod
    def save(self, order):
        pass

class DatabaseStorage(Storage):
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        def save(self, order):
            print(f"Order {order.order_id} ({order.get_order_type()}) saved
to Database")

class FileStorage(Storage):

    def save(self, order):
        print(f"Order {order.order_id} ({order.get_order_type()}) saved
to File")

# -----
# 5. ORDER SERVICE
# -----

class OrderService:

    def __init__(self, payment_method, notification, storage):

        self.payment_method = payment_method
        self.notification = notification
        self.storage = storage

    def place_order(self, order):

        total_amount = order.get_total()

        # Payment
        self.payment_method.pay(total_amount)

        # Save Order
        self.storage.save(order)

        # Notification
        self.notification.send(
            f"{order.get_order_type()} with ID {order.order_id} placed
successfully. Amount: {total_amount}"
        )

# -----
# 6. MAIN PROGRAM WITH USER INPUT

```

```
# -----  
  
if __name__ == "__main__":  
  
    # User Inputs  
    order_id = int(input("Enter Order ID: "))  
    amount = float(input("Enter Order Amount: "))  
  
    print("\nSelect Order Type")  
    print("1. Regular Order")  
    print("2. Discounted Order")  
    print("3. Priority Order")  
  
    order_choice = int(input("Enter choice: "))  
  
    # Create Order  
    if order_choice == 1:  
        order = RegularOrder(order_id, amount)  
  
    elif order_choice == 2:  
        discount = float(input("Enter Discount Amount: "))  
        order = DiscountedOrder(order_id, amount, discount)  
  
    elif order_choice == 3:  
        priority_fee = float(input("Enter Priority Fee: "))  
        order = PriorityOrder(order_id, amount, priority_fee)  
  
    else:  
        print("Invalid Order Type")  
        exit()  
  
    # Payment Method  
    print("\nSelect Payment Method")  
    print("1. Credit Card")  
    print("2. UPI")  
    print("3. Wallet")  
  
    payment_choice = int(input("Enter choice: "))  
  
    if payment_choice == 1:  
        payment = CreditCardPayment()  
  
    elif payment_choice == 2:
```

```
        payment = UPIPayment()

    elif payment_choice == 3:
        payment = WalletPayment()

    else:
        print("Invalid Payment Method")
        exit()

    # Notification Method
    print("\nSelect Notification Method")
    print("1. Email")
    print("2. SMS")
    print("3. Push Notification")

    notification_choice = int(input("Enter choice: "))

    if notification_choice == 1:
        notification = EmailNotification()

    elif notification_choice == 2:
        notification = SMSNotification()

    elif notification_choice == 3:
        notification = PushNotification()

    else:
        print("Invalid Notification Method")
        exit()

    # Storage Method
    print("\nSelect Storage Method")
    print("1. Database")
    print("2. File")

    storage_choice = int(input("Enter choice: "))

    if storage_choice == 1:
        storage = DatabaseStorage()

    elif storage_choice == 2:
        storage = FileStorage()
```

```
else:
    print("Invalid Storage Method")
    exit()

# Create Service
service = OrderService(
    payment,
    notification,
    storage
)

# Place Order
print("\n----- ORDER PROCESSING -----")
service.place_order(order)
```

OUTPUT:

```
PS C:\Users\prerana\OneDrive\Desktop\PYTHON> python order__system.py
Enter Order ID: 789
Enter Order Amount: 800

Select Order Type
1. Regular Order
2. Discounted Order
3. Priority Order
Enter choice: 3
Enter Priority Fee: 400

Select Payment Method
1. Credit Card
2. UPI
3. Wallet
Enter choice: 2

Select Notification Method
1. Email
2. SMS
3. Push Notification
Enter choice: 1

Select Storage Method
1. Database
2. File
Enter choice: 1

----- ORDER PROCESSING -----
Paid 1200.0 using UPI
Order 789 (Priority Order) saved to Database
Email sent: Priority Order with ID 789 placed successfully. Amount: 1200.0
PS C:\Users\prerana\OneDrive\Desktop\PYTHON> 
```